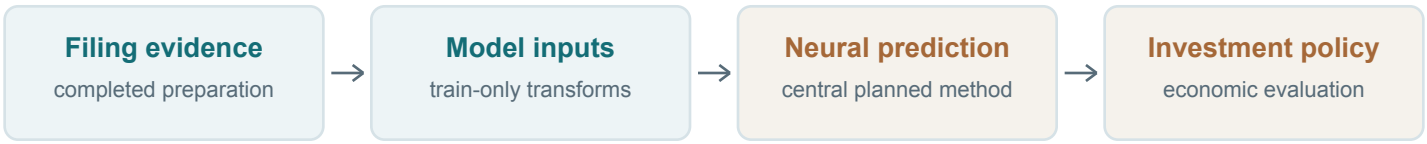
Machine learning for 13D investment selection

Advanced Machine Learning | Discussion 13 October 2026 | Evidence through 12 October

Research question: can a small neural network predict which initial 13D disclosures lead to stronger subsequent benchmark-relative returns, and does LLM-extracted information improve that prediction?

Current status: a historical following baseline, a candidate feature table and chronological preprocessing are complete. Predictive fitting is next. The class contribution is a controlled comparison of supervised methods and information sets; a negative result remains informative.



Central model comparison

Method	Purpose in the experiment
Training-mean prediction	Naive reference: predict the mean outcome available in the training set.
Regularized regression	Transparent linear comparison with penalties that limit model flexibility.
Gradient-boosted trees	Nonlinear comparison that can capture interactions among inputs.
Small feedforward neural network	Central method: learn nonlinear combinations of filing and company inputs. Architecture and tuning budget still to freeze.

Prediction target and investment decisions

The proposed first target is **504-session cumulative stock return minus its assigned historical Russell size-segment ETF-proxy return**. An illustrative stock return of 30% versus a 10% proxy gives a target of +20 percentage points. This is regression, not an invest/not-invest label.

Decision	First experiment	Later extension
Whether to invest	Predict relative return; freeze a selection rule	Utility- or threshold-based policy
When to enter	Existing delayed-entry convention	Compare explicit entry delays
Holding period	Two years proposed	Compare additional horizons
Position size	Existing 5% targets and constraints	Use return/risk estimates for allocation

Historical motivation: simple following has not established an investable edge. Across 56 focused two-year events, mean gross Russell-relative excess is +8.66 points, but median excess is -14.91 and the 95% mean interval spans -17.34 to +40.43. Exploratory 2018-2025 research, not an ML result. [E1]

Data and the broader modeling sample

The first model includes the broader eligible, covered 13D universe. Activist candidates are a diagnostic subgroup; an unresolved identity does not establish that a filer is a non-hedge-fund or non-activist investor.

Two preparation paths with different units

Path	Completed scope	Interpretation
Filing facts	75,631 archive records; 15,813 initial-filing candidates processed and audited, 2013-2025	Facts and prior histories; not 15,813 independent investments. [E2]
Return-labelled events	2,107 scheduled two-year events; 694 calculable stock/proxy labels; 667 single-filing feature joins	Separate scheduling and coverage checks; entry years 2018-2023, follow-up through 2025. [E3-E4]

The 27 remaining calculable events combine multiple filings and need a feature-aggregation rule. Most of the other scheduled events lack a usable Russell assignment. The covered subset may differ from the excluded population.

Disjoint groups inside the 667-event feature table

Candidate identity group	Events	Use in this experiment
Focused activist	55	Report subgroup results; membership remains a candidate classification.
Other Event-Driven	12	Same broader input preparation.
Other HFR	44	Same broader input preparation.
Other / unresolved	556	Retain observations and identity uncertainty.

Ten source-based candidate fields

Information set	Fields and limits
Company context | 2	Security-level market capitalization from a prior snapshot; pre-disclosure industry.
Ownership / extraction | 3	Parsed reporting-person count, largest valid reported percentage, percentage-field coverage. Holdings are not summed.
Disclosure timing | 1	Calendar lag between cover event and filing date; not a legal-deadline determination.
Prior activity | 4	Header reporter count and strictly prior submission/issuer/reporter histories; CIK proxies, not manager skill.

557/667 events have an ownership-percentage summary; 110 remain missing, including 14 unresolved security-attribution cases. Current HFR labels and retrospective activist classifications are not model inputs. Liquidity, volatility, momentum and fundamentals remain to be assembled. [E4]

Chronological preparation and evaluation

A two-year return label enters training only after its outcome window ends. Each prediction year's preprocessing uses earlier available examples. Future inputs do not reset the fill values, numerical scales or industry vocabulary.

Completed development input tables

Prediction year	Earlier training events	Later events	Encoded inputs
2021	118	106	51
2022	254	110	63
2023	365	86	69

Training sets overlap and their counts must not be added. Later observations are previously examined development history, not an untouched final holdout. Inner tuning folds still need label-maturity and dependence checks. [E5]

What the preprocessing does

Method	Implemented rule
Log(1+x)	Compress the numerical spread of company size and count/lag inputs while preserving zero.
Median imputation	Fill missing numerical inputs using observed training values; retain missing flags.
Standardization	Subtract the training mean and divide by its population standard deviation.
Industry encoding	One binary column per training category; unseen later industries use an all-zero category vector and are recorded.
Constant-column mask	Remove columns with no training variation; apply the same saved selection later.

In the 2023 fold, the missing ownership fill is 8.65%, a computational placeholder rather than a factual estimate. Ten original fields become 69 retained inputs, including 58 industry columns. Seven of 86 later events have unseen industries. These dimensions and sparse categories argue for bounded model capacity. [E5]

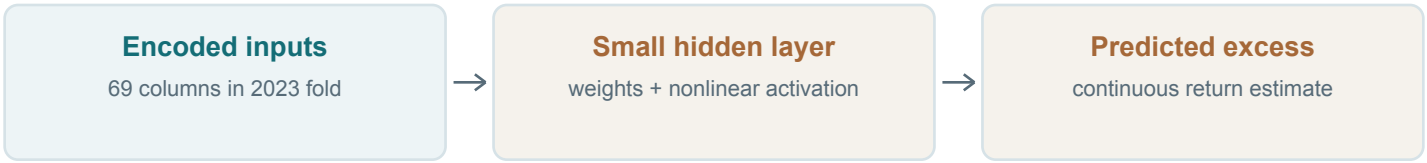
Proposed evaluation and interpretation

Measure	What it asks
Prediction error	Mean absolute error and root mean squared error versus a training-mean reference.
Ranking usefulness	Do higher predictions identify stronger subsequent outcomes? Report rank correlation and subgroup coverage.
Economic relevance	Apply a frozen selection rule under comparable entry, benchmark, 5% sizing, capacity, costs and cash conventions.

Source fingerprints, exact quotations, identifier joins and dates pass local checks; preprocessing replays exactly. These checks do not certify semantic extraction. Historical vendor publication timing is unverified, and corporate-action accounting limits affected portfolio comparisons.

Neural network and incremental LLM experiments

A small neural network is the central planned method



A feedforward network learns weighted combinations of inputs. A nonlinear activation lets those combinations interact; the output estimates relative return. Freeze architecture, loss, regularization, stopping and a modest tuning budget before results. Report variability across training seeds. [M1]

Separate information sets within the same model families

Experiment	Inputs	Question
A | deterministic	Current structured fields and verified additions	Can supervised selection improve on the references?
B | assisted attribution	A plus validated LLM ownership/identity interpretation	Does better attribution improve prediction?
C | disclosed intentions	B plus validated Item 4 objectives	Do stated plans add information beyond the other inputs?

Code deterministically parses, checks citations and replays stored outputs. Fresh LLM interpretation can vary: evaluate a bounded repeated-extraction sample under frozen prompts/settings and retain uncertainty. Agreement measures stability, not correctness or calibrated probability. LLM extraction variability is distinct from network training-seed variability.

Human development references must precede model answers on eligible review cases; assisted practice remains separate. Compare common samples and equal tuning budgets, report coverage changes, and validate extraction before broad feature augmentation. No validated intent accuracy or incremental result exists yet.

Next modeling checkpoints

1. Freeze the target, available features and outcome-mature evaluation; establish naive and regularized comparisons. **2.** Fit the small network and boosted-tree comparison with honest chronological development reporting. **3.** Add a validated, bounded LLM feature experiment, then assess after-cost investment relevance.

A credible unused final sample requires an exposure audit; otherwise final validation should be prospective. A new two-year prospective result extends beyond the course timeline. No prospective collection has begun.

Evidence and research status

Record	Supporting evidence
E1 | 24 September	Frozen historical following results and long-horizon trading conventions.
E2 | 24-25 September	Completed factual staging and audit: 15,813 initial-filing candidates.
E3-E5 | 12 October	Sample feasibility, feature/source joins and train-only preprocessing; exact local replay.

M1: scikit-learn neural-network documentation; M2: chronological validation documentation. Local evidence index accompanies the brief.

AI-assisted research tooling and writing. No predictive model, validated intent features or incremental investment result has been produced. Existing human-review records and historical outputs are preserved.